

Microcontroller Registers Overview (STM32 / Cortex-M0)

Definition: Registers are small, fast storage locations inside the microcontroller used for execution, control, and communication with hardware.

1. CPU (Core) Registers

Used by the processor to execute instructions and manage program flow.

Includes: R0–R12 (general purpose), MSP/PSP (stack pointers), PC (program counter), LR (link register), xPSR (status), CONTROL, PRIMASK.

2. Peripheral Registers

Used to control hardware peripherals such as GPIO, UART, TIMERS, ADC.

Examples: GPIOA->MODER, RCC->AHBENR, USART1->CR1.

3. System Control Registers

Used for managing system-level operations like interrupts and core configuration.

Includes NVIC, SysTick, and SCB registers.

MSP (Main Stack Pointer)

MSP holds the address of the top of the stack in SRAM. It is used by the CPU to store temporary data during function calls and interrupts.

Key Insight

CPU registers handle execution, peripheral registers control hardware, and system registers manage global behavior.